RATING METHODOLOGY

8 April 2024

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Rating Methodology

# Enhanced Equipment Trust Certificates

This rating methodology replaces the *Enhanced Equipment Trust and Equipment Trust Certificates Methodology* published in July 2018. While this methodology reflects many of the same core principles as the 2018 methodology, we have introduced a scorecard and added more detail regarding the key rating factors and other rating considerations that may be important in rating enhanced equipment trust certificates and equipment trust certificates. We have also made editorial changes to enhance readability.

## Scope

This methodology applies to collateralized financings globally of aircraft, aircraft engines or parts, and rail equipment (rolling stock) that are funded by equipment payments made by a single underlying corporate obligor. These transactions are structured such that, if the obligor makes the required payments due under the transaction agreements and the related payments for administrative expenses provided for in the transaction documentation, there will be sufficient funds available to meet scheduled principal, interest and fees on the debt financing. These equipment obligations are subject to laws that provide for ongoing payment, or for the return of the collateral if the underlying obligor does not affirm the notes in the event of the initiation of a bankruptcy or insolvency.

Financings that we rate using this methodology are typically enhanced equipment trust certificates (EETCs) for financings of specific aircraft or aircraft engines or parts, and equipment trust certificates (ETCs) for rolling stock. The collateral typically does not change over time and is the sole source of security under a certificate default scenario.

Financings that are backed by pools of assets that rely on cash flows from multiple obligors, usually have cash flow waterfalls with variable debt service distributions, and have the ability to transition assets to a different operator at lease expiry or following an incumbent operator default, are rated using other methodologies, for example our methodologies for rating securities backed by aircraft and associated leases, equipment lease and loan securitizations and railcar leasing securitizations surveillance.

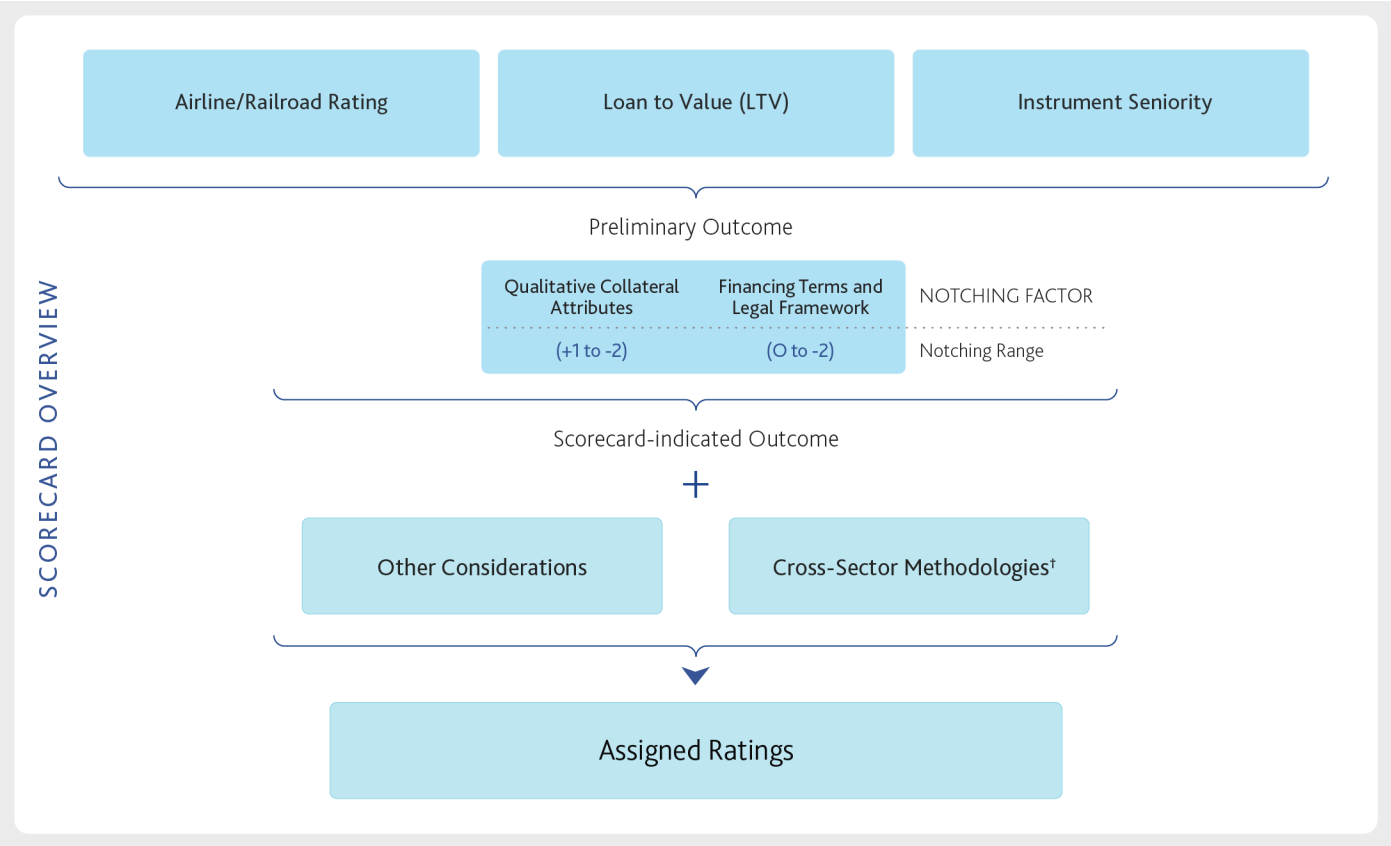
## Rating approach

In this rating methodology, we explain our general approach to assessing credit risk of EETCs and ETCs, including the qualitative and quantitative factors that are likely to affect rating outcomes for these instruments. We use the term certificate in this methodology to refer to an EETC or an ETC. We seek to incorporate all material credit considerations in our ratings

and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

The following schematic illustrates our general framework for the analysis of certificates, which includes the use of a scorecard. The scorecard-indicated outcome is not expected to match the actual rating for each instrument. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1
**Illustration of the EETC and ETC methodology framework**




† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.
*Source: Moody's Ratings*

Our approach considers the extent of rating uplift from an airline or railroad's corporate family rating (CFR) for speculative-grade obligors or, for investment-grade obligors, uplift from the issuer rating, or, if there is no issuer rating, the senior unsecured rating. The uplift reflects the credit protective benefits typically included in certificate financings, which may include: contractual and legal provisions that provide creditors timely access to collateral following an airline or railroad not affirming the equipment notes (e.g., the US Bankruptcy Code, including Section 1110 thereof, or the Cape Town Convention[1]); over-collateralization; liquidity facilities sized to pay at least the next 18 months' interest; and cross-default, cross-collateralization and cross-subordination established by an intercreditor agreement.

We establish a preliminary outcome for each class of debt, which incorporates upward notching, derived from our loan to value (LTV) grids using our estimate of peak LTV, the seniority of the instrument, and the airline or railroad's reference rating.

This publication does not announce a credit rating action. For any credit ratings referenced in this publication, please see the issuer/deal page on https://ratings.moodys.com for the most updated credit rating action information and rating history.

We may adjust the preliminary outcome upward or downward based on our assessment of qualitative collateral attributes such as our opinion of the importance of the collateral to an operator's network, the age of the assets, the size of the global operating fleet, and the number of operators.

We may also assign downward notching to the preliminary outcome based on our assessment of the transaction financing terms, including our assessment of the applicable legal framework, the cross-default, cross-collateralization or cross-subordination features, liquidity facilities, and lessor financing arrangements.

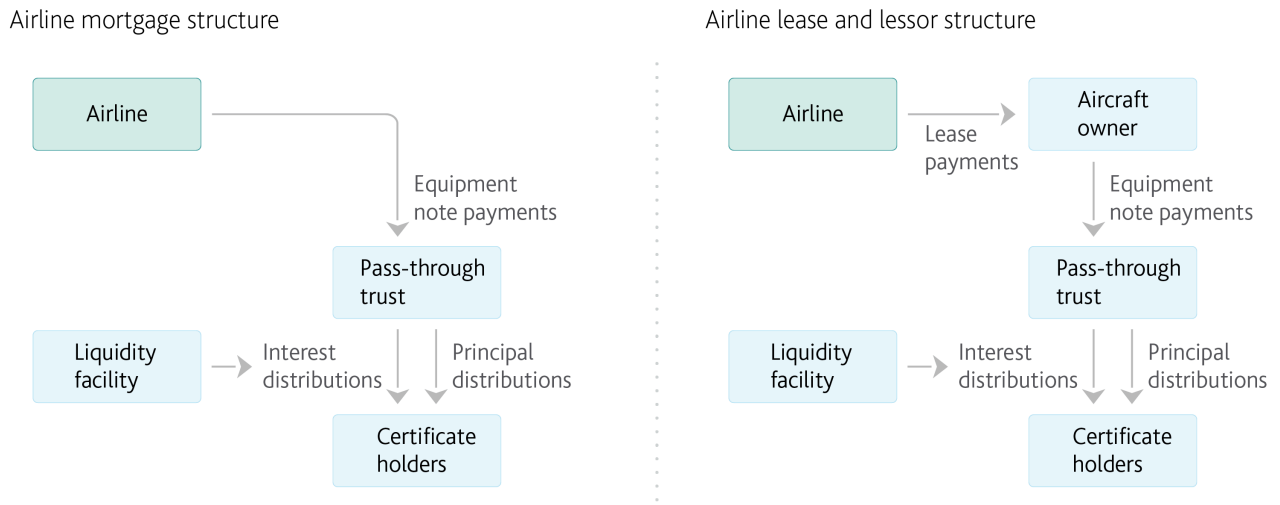
The notching uplift for EETCs is generally higher than for ETCs with similar LTV ratios because EETCs typically benefit from several characteristics that are not present in ETCs, including: a liquidity facility, tranching, lower LTV ratios for senior tranches, cross-default and cross-collateralization features and cross-subordination. There is no tranching in ETCs and their initial LTVs are typically between 75% and 85%, which is higher than the typical initial LTVs on senior tranches of EETCs. ETCs have been issued infrequently since 2005.

## Sector overview

The structures of certificates in EETC and ETC transactions are similar to each other in that each is a pass-through financing, whereby a trust is created to issue the rated instrument (i.e., the certificates), with the proceeds from the sale used to purchase equipment notes issued by an airline or railroad, or by a third-party lessor owner of the collateral.

EETCs arranged by an airline are typically structured as either mortgage or lease financings. In some cases, EETCs are arranged by lessors, which we describe as lessor EETCs in this methodology. For all airline EETCs, there are typically separate equipment notes for each aircraft. In a transaction that includes a lease structure, there is typically a separate lease for each aircraft in addition to the equipment notes. There is typically only one or a few master notes or leases for rail ETCs and lessor EETC transactions. The pass-through trusts receive the contractual payments of interest and principal due on the equipment notes, which in the case of a third-party owner are funded via lease payments received from the airline or railroad that operates the equipment. The cash flows that the trust receives fund the scheduled distributions of interest and pool balances (principal) required by the terms of the certificates.

Exhibit 2
**Typical EETC structures**




*Source: Moody's Ratings*

We typically consider the default probability of certificates to be materially lower than the default probability of the underlying obligor's corporate debt instruments. Factors that help lower the default probability include: the bankruptcy remoteness of the trust created to issue the certificates, the importance of the equipment collateral to the underlying obligor's operations, the need for the underlying obligor to affirm the equipment notes following a bankruptcy or insolvency in order to retain access to the equipment

collateral, the existence of cross-default, cross-collateralization and cross-subordination provisions, and the existence of liquidity facilities.

Certificates rated using this methodology are not direct debt obligations of, nor are they guaranteed by, the obligor (the underlying airline or railroad). As long as the obligor makes the required payments due under the equipment note indentures, conditional sale agreements or leases, and makes the required payments for the administrative expenses provided for in the transaction documentation, the trusts will have funds available to meet scheduled distributions on their certificates.

In the event of the bankruptcy or insolvency of the obligor of the underlying instrument, the obligor could decide not to affirm the notes. However, the obligor would need to do so within the time frame specified by the applicable legal regime (e.g., for aircraft, Section 1110 of the US Bankruptcy Code or the Cape Town Convention, or another legal regime for aircraft; or, for US railroads, Section 1168 of the US Bankruptcy Code) whether it will continue to utilize the equipment financed by the certificate. As implemented by most adopters of the Cape Town Convention, this time frame is typically 60 days from the initiation of the bankruptcy or insolvency, the same as with Section 1110 or Section 1168 of the US Bankruptcy Code. If the obligor elects to continue using the equipment, it would need to make the scheduled interest and principal payments on the equipment notes or leases supporting the certificates during the reorganization. Any past due amounts are also required to be repaid once the affirmation election is made.

In scenarios where the underlying obligor decides not to affirm the notes and the collateral is repossessed, to the extent that the proceeds realized from liquidation of the collateral are insufficient to extinguish the aggregate of fees and expenses, costs to sell, liquidity facility provider claims, accrued interest and the outstanding balance of the underlying financing obligation, the pass-through trusts would have residual unsecured claims against the bankruptcy/insolvency estate of the underlying obligor.

## EETC and ETC scorecard

For general information about how we use the scorecard and for a discussion of scorecard mechanics, please see the "Using the scorecard to arrive at a scorecard-indicated outcome" section. The scorecard does not include or address every factor that a rating committee may consider in assigning ratings in this sector. Please see the "Other considerations" and "Limitations" sections.

Exhibit 3

**EETC and ETC scorecard**

| | Scoring Range |
|---|---|
| **Factor: Airline/Railroad Rating** | |
| Airline/Railroad Reference Rating | Aaa to Ca rating |
| **Factor: Loan to Value (LTV) / Instrument Seniority** | |
| Loan to Value (LTV) | Less than 45% to higher than 100% |
| Instrument Seniority[1] | Senior, First Junior or Second Junior |
| **Notching Factors** | |
| Qualitative collateral attributes | Up to one notch higher or two notches lower |
| Financing terms and legal framework | Up to two notches lower |

[1] Instrument Seniority does not apply to ETCs.
*Source: Moody's Ratings*

## Discussion of the scorecard factors

In this section, we explain our general approach for scoring each scorecard factor, and we describe why it is meaningful as a credit indicator.

### Factor: Airline/Railroad Rating

**Why it matters**

The airline or railroad rating is crucial to our assessment of an EETC or ETC because cash flows from the underlying airline or railroad obligor are the sole source of payments to the pass-through trusts that fund distributions on the certificates (other than the proceeds after disaffirmation of the equipment notes).

The reorganization strategy of a single underlying obligor influences its decisions regarding the types of aircraft it wants to continue to operate, which can lead to a decision to disaffirm the notes, requiring the trustee to monetize the collateral in order to satisfy the obligations. In these scenarios, the then-current market conditions would affect the timing of recoveries and ultimate recovery rates.

**How we assess it**

The reference rating is the airline or railroad's CFR for speculative-grade issuers. For investment-grade issuers, the reference rating is the issuer rating or, if there is no issuer rating, the senior unsecured rating.

### Factor: Loan to Value (LTV) / Instrument Seniority

**Why it matters**

The LTV ratio provides very important indications of debt leverage for the transaction and capacity to repay debt from collateral proceeds when the notes are disaffirmed following a bankruptcy.

The seniority of an EETC tranche determines the priority of debt service payment of that tranche relative to other tranches in the same financing. EETC financings can include one tranche to three or more tranches.

**How we assess it for EETCs**

The LTV notching grids in Exhibits 4-6 show the upward LTV notching for airline EETCs. The numerator in an LTV ratio is the outstanding debt balance and the denominator is our market value estimate. The outstanding debt balance is the total of the outstanding debt balance of the relevant tranche and the outstanding debt balance of all higher ranking tranches. We use the senior notching grid for the most senior tranche in an EETC financing, the first junior notching grid for the next-ranking tranche, and the second junior notching grid for any lower-ranking tranches. The LTV used in scoring this factor is the peak LTV projected over the remaining life of the transaction.

Exhibit 4

## Senior EETC tranche notching grid

| | Airline Rating | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Peak LTV | Aaa | Aa1 | Aa2 | Aa3 | A1 | A2 | A3 | Baa1 | Baa2 | Baa3 | Ba1 | Ba2 | Ba3 | B1 | B2 | B3 | Caa1 or lower |
| < 45% | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 8 | 8 | 9 | 9 | 9 | 9 | 9 | 9 |
| 45% - 50% | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 6 | 7 | 7 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 50% - 55% | 0 | 1 | 2 | 3 | 4 | 5 | 5 | 6 | 6 | 7 | 7 | 7 | 8 | 8 | 8 | 8 | 8 |
| 55% - 60% | 0 | 1 | 2 | 3 | 4 | 4 | 5 | 5 | 6 | 6 | 6 | 7 | 7 | 7 | 7 | 7 | 7 |
| 60% - 65% | 0 | 1 | 2 | 3 | 4 | 4 | 4 | 5 | 5 | 6 | 6 | 7 | 7 | 7 | 7 | 7 | 7 |
| 65% - 70% | 0 | 1 | 2 | 3 | 3 | 3 | 4 | 4 | 5 | 5 | 5 | 6 | 6 | 6 | 6 | 6 | 6 |
| 70% - 75% | 0 | 1 | 2 | 3 | 3 | 3 | 3 | 4 | 4 | 5 | 5 | 5 | 5 | 5 | 6 | 6 | 6 |
| 75% - 80% | 0 | 1 | 2 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 5 | 5 | 5 | 5 | 5 |
| 80% - 85% | 0 | 1 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 85% - 90% | 0 | 1 | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 90% - 95% | 0 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 95% - 100 % | 0 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| ≥ 100% | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

*Source: Moody's Ratings*

Exhibit 5

## First junior EETC tranche notching grid

| | Airline Rating | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Peak LTV | Aaa | Aa1 | Aa2 | Aa3 | A1 | A2 | A3 | Baa1 | Baa2 | Baa3 | Ba1 | Ba2 | Ba3 | B1 | B2 | B3 | Caa1 or lower |
| < 45% | 0 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 7 | 7 | 8 | 8 | 8 | 8 | 8 | 8 |
| 45% - 50% | 0 | 0 | 1 | 2 | 3 | 4 | 5 | 5 | 6 | 6 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 50% - 55% | 0 | 0 | 1 | 2 | 3 | 4 | 4 | 5 | 5 | 6 | 6 | 6 | 7 | 7 | 7 | 7 | 7 |
| 55% - 60% | 0 | 0 | 1 | 2 | 3 | 3 | 4 | 4 | 5 | 5 | 5 | 6 | 6 | 6 | 6 | 6 | 6 |
| 60% - 65% | 0 | 0 | 1 | 2 | 3 | 3 | 3 | 4 | 4 | 5 | 5 | 6 | 6 | 6 | 6 | 6 | 6 |
| 65% - 70% | 0 | 0 | 1 | 2 | 2 | 2 | 3 | 3 | 4 | 4 | 4 | 5 | 5 | 5 | 5 | 5 | 5 |
| 70% - 75% | 0 | 0 | 1 | 2 | 2 | 2 | 2 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 5 | 5 | 5 |
| 75% - 80% | 0 | 0 | 1 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 |
| 80% - 85% | 0 | 0 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 85% - 90% | 0 | 0 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 90% - 95% | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 95% - 100 % | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| ≥ 100% | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

*Source: Moody's Ratings*

Exhibit 6

**Second junior EETC tranche notching grid**

| | Airline Rating | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Peak LTV | Aaa | Aa1 | Aa2 | Aa3 | A1 | A2 | A3 | Baa1 | Baa2 | Baa3 | Ba1 | Ba2 | Ba3 | B1 | B2 | B3 | Caa1 or lower |
| < 45% | 0 | 0 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 6 | 6 | 7 | 7 | 7 | 7 | 7 | 7 |
| 45% - 50% | 0 | 0 | 0 | 1 | 2 | 3 | 4 | 4 | 5 | 5 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 50% - 55% | 0 | 0 | 0 | 1 | 2 | 3 | 3 | 4 | 4 | 5 | 5 | 5 | 6 | 6 | 6 | 6 | 6 |
| 55% - 60% | 0 | 0 | 0 | 1 | 2 | 2 | 3 | 3 | 4 | 4 | 4 | 5 | 5 | 5 | 5 | 5 | 5 |
| 60% - 65% | 0 | 0 | 0 | 1 | 2 | 2 | 2 | 3 | 3 | 4 | 4 | 5 | 5 | 5 | 5 | 5 | 5 |
| 65% - 70% | 0 | 0 | 0 | 1 | 1 | 1 | 2 | 2 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 4 |
| 70% - 75% | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 2 | 2 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 |
| 75% - 80% | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 3 |
| 80% - 85% | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 85% - 90% | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 90% - 95% | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 95% - 100 % | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| ≥ 100% | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

*Source: Moody's Ratings*

In our initial rating assignments and ongoing monitoring of EETCs, the primary sources of aircraft values in our peak LTV projections are estimates from independent International Society of Transport Aircraft Trading (ISTAT) certified appraiser valuation guides. The offering documentation typically includes desktop appraisals from three ISTAT-certified appraisers for each aircraft in the transaction. These appraisals serve as a reference point for our LTV analysis at initial rating assignment.

We estimate peak LTV values using our estimates of current market values that assume the aircraft are at their maintenance half-lives, except for new deliveries, where we use full-life values. We do not adjust our estimate of market values to reflect the current maintenance status of the aircraft. Offering memoranda LTVs are generally based on the lower of the mean or median of the appraised maintenance-adjusted base values, i.e., economic value of an aircraft in an open, unrestricted market with balanced supply and demand, provided by three appraisal firms. Base value typically assumes average physical condition and relies on historical value trends normalized for swings in economic conditions. Our LTV ratio estimates are derived from market values rather than base values because base values assume a balance of industry supply and demand that can significantly differ from current market conditions.

We calculate the LTV at each scheduled payment date to identify the peak LTV over the transaction's term. We generally estimate the LTV without an assumed full draw of the liquidity facilities and incurrence of costs to sell, the latter of which we assume to be 5% of market value. Where the airline is closer to default, e.g., Caa rated, we include these priority claims in our LTV estimates. On a combined basis, these two priority claims have historically represented 8% to 15% of outstanding debt, on average, reducing the excess collateralization for certificate holders. The low end of this range reflects issuance during periods of low interest rates, and thus low fixed interest rates on the EETCs. The consideration of priority claims becomes more prominent for EETCs of airlines rated at the lower end of the rating scale because the probability of an underlying issuer's bankruptcy or insolvency may increase. These priority claims reduce the excess collateralization for each tranche.

When projecting market values of newly delivered aircraft, we typically reduce values of all models by 7.5% in the first year. For year 2 and thereafter, we typically reduce values of narrow-body and regional passenger aircraft by either 4% or 5%, and of wide-body and freighter aircraft by either 5% or 6%. The lower rates are applied to models that are currently in production. We temper the annual rates of decline with a conservative assumption for annual inflation. We may change our assumptions for the annual rates of market value decline or inflation if, for example, there was a sustained trend of aircraft value retention being higher or lower than our current assumptions.

We apply a conservative assumption of zero residual value for all aircraft when projecting market values. Since peak LTV often occurs within a year of delivery, our assumption of zero residual value does not significantly affect LTV scores or associated notching, but

it may result in modestly smaller increases in the projected excess collateralization over the tenor of the EETC, and it provides a better alignment with how aircraft values have trended over time. Using our higher annual rates of decline in market value for prior-generation aircraft and our conservative inflation assumption can result in projected market values near the end of a transaction's life that are lower than residual values in the market for a particular aircraft. This is because the part-out value of an aircraft, primarily its engine components, typically sets a floor for the aircraft's market value toward the latter part of its useful life as a passenger aircraft.

As part of our monitoring of aircraft EETC transactions, we monitor LTVs, and our valuation estimates are informed by our aircraft appraisers' opinions of current market values. The aircraft valuation data we use are typically updated semiannually from at least two ISTAT-certified aircraft appraisal firms, and may be updated more frequently. We also gather information about aircraft values and trends from other sources.

Aircraft values can be volatile through economic cycles, which can result in lower market values and higher LTVs during cyclical troughs. Because our ratings are forward-looking, the long-term fundamental outlook for aircraft values is generally a more important driver than minor or temporary changes in these values.

Our LTV notching grids highlight a greater emphasis on collateral value as airline credit quality weakens. The larger notching at the lower end of the corporate ratings scale recognizes the greater weight we place on the importance of the collateral to the airline, the probability of affirmation of a transaction should a bankruptcy or insolvency proceeding take place, and recovery prospects if equipment notes are not affirmed following a bankruptcy or insolvency proceeding.

**How we assess it for ETCs**

The LTV notching grid in Exhibit 7 shows the upward notch scoring for LTV for rail ETCs.

Exhibit 7

**ETC notching grid**

| | Railroad Rating | | | | | | |
|---|---|---|---|---|---|---|---|
| LTV | Aaa | Aa | A | Baa | Ba | B | Caa or lower |
| < 50% | 0 | 2 | 4 | 4 | 4 | 4 | 4 |
| 50% - 80% | 0 | 1 | 3 | 4 | 4 | 4 | 4 |
| 80% - 90% | 0 | 1 | 3 | 4 | 4 | 4 | 4 |
| 90% - 100% | 0 | 1 | 3 | 3 | 3 | 3 | 3 |
| ≥ 100% | 0 | 1 | 1 | 1 | 1 | 1 | 1 |

*Source: Moody's Ratings*

The numerator is the outstanding debt balance and the denominator is our valuation estimate. Market values for railroad rolling stock, whether freight cars or locomotives, are not readily available. For rail ETCs, we estimate asset values at the issuance date of a transaction, typically based on invoice prices for new equipment. ETC LTVs are typically assumed to be constant over the life of the transaction based on an assumption that the decline in value over time is proportional to the amortization profile of the transaction. We may adjust our basis for estimating asset valuation for an ETC where we consider this assumption to be inappropriate (e.g., where the transaction tenor or amortization is non-standard). We also may revise our valuation estimate lower if market data or other information indicates that our valuation estimate is higher than market value.

## Notching factors

The scorecard contains two notching factors. Our assessment of the notching factors may result in upward or downward adjustments to the preliminary outcome. We apply these adjustments in whole-notch increments, with a maximum of four alphanumeric notches down and a maximum of one alphanumeric notch up from the preliminary outcome to arrive at the scorecard-indicated outcome.

In cases where we consider that the credit weakness or credit strength represented by a notching factor, or by these factors in aggregate, is greater than the scorecard range, we incorporate this view into the rating, which may be different from the scorecard-indicated outcome. For a discussion of scorecard mechanics, see the "Using the scorecard to arrive at a scorecard-indicated outcome" section.

Exhibit 8
**Notching factors**

| Notching factor | Notching range |
|---|---|
| Qualitative collateral attributes | +1 to -2 |
| Financing terms and legal framework | 0 to -2 |

*Source: Moody's Ratings*

## Notching factor: Qualitative collateral attributes

**Why it matters**

Qualitative features of the collateral can provide important indications of the likelihood that an airline would affirm the notes in a bankruptcy or insolvency proceeding. The importance to future operations of specific aircraft models that serve as collateral for EETCs can raise or lower the probability of default meaningfully for such EETCs in an airline's bankruptcy or insolvency.

The size of the global fleet and number of operators, i.e., potential buyers or lessees, can also provide additional indications of the forward trajectory of collateral value.

In our assessment of rail ETCs, qualitative collateral attributes are generally less relevant because of their interoperability among rail operators.

**How we assess it**

In assessing the likelihood of affirmation of equipment notes (or leases) related to rated EETCs in a scenario of an airline bankruptcy or insolvency, we consider the essentiality of the collateral to the airline's strategy and network.

We generally expect that an airline would reorganize in an insolvency scenario. As opposed to a liquidation, a reorganization preserves the possibility of a carrier affirming its obligations under most or all of its equipment notes during the period provided by the applicable legal regime.

Even in a reorganization, an airline could have a smaller route network requiring fewer aircraft, which could affect an airline's decision to affirm the equipment notes in one or more of its EETCs. Also, an EETC collateralized by a small number of aircraft or by older or less cost-effective aircraft in the carrier's fleet would have a higher probability of disaffirmation, should a bankruptcy occur.

We consider several qualitative attributes of the aircraft collateral in assessing the likelihood of affirmation, in addition to the impact of these qualitative attributes on our market value estimates, including those in Exhibit 9.

Exhibit 9
**Qualitative aircraft attributes that we consider in assessing EETCs**

| Collateral attribute | Comment |
|---|---|
| Importance of the aircraft in the issuer's fleet and to its network and strategy | Transactions involving aircraft likely to be unnecessary or less important to a scaled-down network under a reorganization scenario, or by aircraft approaching or older than 20 years that are likely to be replaced by younger vintages, could be notched lower than transactions for the same airline involving aircraft that we believe will remain core to the airline's network over the EETC tenor. |
| Size of the airline industry global operating fleet by model and type<br>Number of operators of a particular model and type | The size of the global fleet and the number of operators of a model or type of aircraft are indicators of potential market demand for an aircraft model or type over time. Transactions involving aircraft that are part of relatively small global fleets or have relatively few operators could be notched lower than transactions for the same airline involving aircraft that are part of large global fleets operated by many airlines. |

*Source: Moody's Ratings*

Airlines' fleets and the product offerings of original equipment manufacturers have become more concentrated over time. Fleet streamlining has made airline operations more efficient and has reduced the number of marginal, or excess, aircraft in their fleets. This change in fleet composition results in fewer marginal aircraft types in airline fleets.

Obligations collateralized entirely with younger vintages of prior-generation aircraft will likely receive no upward or downward notching for this notching factor, as will EETCs with the current generation of in-production aircraft where the transaction tenor

remains well within the expected economic lives of these models and where such models are expected to remain integral to the airline's network over the EETC's term.

Downward notching may be applied where the collateral includes older aircraft relative to transactions with similar LTVs or relative to the rest of that airline's fleet. Downward notching may also be applied to an EETC with aircraft for which the size of the global fleet is relatively small or where the number of operators globally is relatively small.

In circumstances where we have extremely high confidence that an EETC will be affirmed in a reorganization scenario, we may assign an upward notch for qualitative collateral considerations.

### Notching factor: Financing terms and legal framework

**Why it matters**

Financing terms are important in our analysis of EETCs and ETCs. The terms of trust documents, indentures of underlying equipment notes, and intercreditor agreements are generally standardized and generally include cross-default, cross-collateralization and cross-subordination, and liquidity facilities, as described in this section. Absence of these features and the presence of nonstandard terms may increase risk to creditors because there may be a greater chance that proceeds realized in a disaffirmation scenario may be insufficient to cover the outstanding obligations. The considerations that allow certificate ratings to exceed the issuer rating or corporate family rating of an airline or railroad depend extensively on the dependability of the legal framework as it relates to bankruptcy and insolvency. Downward notching may result from the absence of standard terms, inclusion of nonstandard contractual provisions or from the documents being subject to a legal jurisdiction that could result in reduced creditor protections relative to other EETC financings.

**How we assess it for EETCs**

*Cross-default*

The cross-default feature may lower the likelihood of a disaffirmation of equipment notes in scenarios where some but not all of the aircraft are important to the airline's post-reorganization strategy. This feature prevents the airline from cherry-picking from the collateral pool and deciding not to affirm notes related to unwanted aircraft. EETCs are supported by a series of underlying equipment notes issued by the underlying obligor airline or third-party owner of each aircraft to the pass-through trusts that issue the EETCs. Each equipment note relates to the mortgage on an individual aircraft. In a lease EETC, each aircraft is held under a separate lease that secures its related equipment note(s). We review the transaction documents for cross-default provisions among the equipment notes and the leases.

*Cross-collateralization*

In the event of a disaffirmation by the airline, cross-collateralization provisions in underlying equipment notes provide potential recovery enhancement for investors, and may even reduce default risk. Surplus proceeds received from the sale of one aircraft are retained by the trust and can compensate for a shortfall on sales of other aircraft. Cross-collateralization is likely to be more beneficial in an EETC that has aircraft of different types or vintage than in an EETC with aircraft of the same type or vintage because of the potential lower correlation of changes in value across aircraft models.

*Cross-subordination*

Cross-subordination is important because it facilitates the tranching in an EETC and helps assure that the creditors will benefit from the stated priority of their claims to the payments received and the collateral. The cross-subordination provisions are typically detailed in a transaction's intercreditor agreement and administered by a subordination agent who receives and routes payments made by the underlying obligor on each class of equipment notes to the respective pass-through trustees for the corresponding class of EETCs. Specifically, the intercreditor agreement defines the waterfall, i.e., the priority of distribution of funds received by the subordination agent on behalf of the pass-through trusts.

Interest and principal paid on the underlying equipment notes are the primary source of cash paid into the pass-through trust(s) for EETCs that are performing pursuant to the distribution schedules in their offering documents. Where all of the notes in an EETC are performing pursuant to the original contractual terms, the application of the expected distributions for transactions issued since 2007 is typically as follows: first, to service provider fees and expenses, which are typically subject to caps; then to interest on the

senior certificates; then to interest on the junior certificates, then to interest on the next more junior class of certificates, if any; then to principal on the senior certificates; then to principal on the junior certificates; then to principal on the next more junior class of certificates, if any. The post-default waterfall is the same as this except that cost of repossession and sale of collateral and liquidity facility provider claims receive distributions immediately after service provider fees and expenses, before certificate holders of any class receive distributions. In a transaction that uses a pay-in-kind (PIK) feature in place of a liquidity facility, the principal on the senior class is distributed before interest and principal on the junior classes.

The cross-subordination provisions become effective upon the default of the underlying financing agreement(s) and if these defaults are not cured by the underlying obligor pursuant to a Section 1110(a) election or under the Cape Town Convention's Alternative A provisions or another applicable legal regime. Liquidity facility drawings and proceeds from the disposal of one or more aircraft following a default on the underlying financing agreements and repossession of the related collateral following an 1110(c) election are the cash flows the pass-through trusts receive for nonperforming EETCs. A section 1110(c) election provides for the airline to disaffirm its obligations due on a financing that underlies an EETC transaction, and it would occur by the end of the 60-day Section 1110 period. The airline is required to return the aircraft to the trustee following its disaffirmation of a financing. The cross-subordination provisions govern the distribution of these cash flows to certificate holders.

Where one or more equipment notes or leases within an EETC is in default, the cross-subordination feature acts to reduce the risk to the senior certificates at the expense of the junior certificates. If the senior and junior classes of the equipment notes of a particular aircraft are repaid in full, then all of the net proceeds, i.e., after fees and disposal costs, received by the subordination agent are directed by it to the repayment of accrued interest and the pool balances of the certificates in the order described above.

*Liquidity facilities*

Liquidity facilities can delay the default of an EETC whose underlying notes have been disaffirmed by an airline that has filed for bankruptcy protection and may provide sufficient time to allow for repayment from collateral before a default occurs. The liquidity facility is usually a revolving line of credit. The size and tenor of the liquidity facility are intended to provide sufficient time for an orderly liquidation of the foreclosed collateral. These facilities are typically sized to meet the next three semiannual or next six quarterly interest payments due on the EETCs following an airline's bankruptcy. Repossession and monetization of the collateral is anticipated to be completed within 18 months of the filing date, avoiding a default for nonpayment of scheduled interest during this period. Skipping payments of scheduled pool balances of outstanding EETCs (principal) is not an event of default for EETCs pursuant to their terms.

The terms of EETCs typically define a default as the nonpayment by the pass-through trustee of a scheduled interest payment or nonpayment of the pool balance (principal) then outstanding at the legal final maturity date. Legal final maturity dates typically lag the scheduled final payment date by the tenor of the liquidity facility, typically 18 months. The legal final distribution date for tranches that lack a liquidity facility is that tranche's scheduled final payment date. A C tranche, either the third or fourth tranche in a transaction, will typically not have the support of a liquidity facility.

The 18-month tenor for liquidity facilities is the standard for transactions governed by Section 1110 and for non-US legal regimes where we expect that creditors would have similar timely access to collateral following an EETC default. To avoid downward notching for this factor, a longer tenor liquidity facility may be needed in transactions for which there is less certainty over timely enforcement or timely repossession of collateral, or where the aircraft models could require a longer collateral monetization period.

EETC terms and conditions typically include a minimum rating requirement for the provider of the liquidity facility. In our view, a financial institution with a long-term rating of Baa2 or a counterparty risk assessment of at least Baa2(cr) will typically be able to fulfill a commitment to fund liquidity facility draws, should a drawing request be made. EETC documentation provides for the timely replacement of the liquidity facility provider if its relevant rating falls below the rating required by the transaction's terms. Otherwise, the facility may be drawn, cash-collateralizing the commitment. Our EETC ratings generally are not capped by the rating of the liquidity provider because of the low likelihood of the airline obligor and the financial institution providing the liquidity facility defaulting at the same time, and because of the liquidity facility replacement provisions. We typically also consider the revolving facility's terms and conditions, including required representations and warranties for drawings.

In some transactions, a PIK feature is included in place of a liquidity facility. In these transactions, scheduled interest payments on the certificates accrue following a missed payment on the equipment notes, rather than being funded currently from draws on liquidity facilities. In a PIK transaction, there is no liquidity facility priority claim. However, the certificate holders' claim increases with each missed scheduled payment. The PIK feature typically runs for an 18-month period, providing a similar time period to realize collateral as in transactions with liquidity facilities. We typically do not assign negative notching under this notching factor for transactions using the PIK feature. However, a downward notch may be applied in transactions with PIK coupons above 5%. In these cases, the incremental claim for accrued interest based on the stated plus the penalty interest rate could sufficiently increase the LTV to warrant a lower rating.

*Lessor EETCs*

Lessor EETCs are arranged by aircraft leasing companies rather than airlines. Typically, transaction structures and terms meaningfully mirror traditional EETC financings. The underlying obligors are trusts created for the benefit of aircraft-owning subsidiaries of the aircraft leasing company. These trusts are funded by a rated airline lessee's payments under one or more operating leases. These lease payments service the interest and principal due on the transaction's equipment notes, which flow through to the certificate holders. The corporate family, issuer rating or senior unsecured rating of the airline lessee is the reference rating to which we apply notching in a lessor EETC.

The lessor EETC-related equipment leases are not cross-defaulted. The default of one or more of the leases is an event of default for all of the notes, but not for the other leases. Lessor EETCs allow for the sale of one or more of the aircraft in the collateral. This feature does not exist in traditional EETCs. For each aircraft sold, if any, the transaction's terms require the issuer to repay 105% of the amount of note debt attributable to the sold aircraft. By contrast, in a typical airline EETC that is cross-defaulted and cross-collateralized, any early payoff would generally require full payoff of the transaction.

Downward notching may also apply to lessor EETC transactions. Terms of lessor EETCs allow for the sale of one or more aircraft in the transaction. Although the terms require the retirement of 105% of the equipment note debt attributable to the sold aircraft, reducing the number of aircraft in the transaction could weaken the effects of the cross-collateralization feature or affect our qualitative assessment of the collateral.

*Legal framework*

The legal framework governing the treatment of aircraft or equipment secured by EETCs provides important indications of the outcomes for creditors following a bankruptcy filing of the obligor. For example, the legal framework governs considerations such as the time frame for the obligor to make decisions regarding whether it will affirm or disaffirm notes, the rights of creditors to realize collateral if interest and principal payments cease or are interrupted, and whether and how priority of claim will be respected. Because the issuers of the certificates are bankruptcy-remote, pass-through trusts that are independent of the obligor of the underlying financings of those trusts, EETCs do not default upon a bankruptcy or insolvency filing by the underlying airline. The provisions of Section 1110 of the US Bankruptcy Code provide for EETC collateral to be exempt from the automatic stay provisions of Section 362 of the US Bankruptcy Code. Provisions of the Cape Town Convention provide similar provisions for EETCs whose underlying financings are subject to non-US legal regimes that have also implemented the Cape Town Convention in their laws. For rail ETCs, the relevant US Bankruptcy Code provisions are contained in Section 1168.

The legal framework for EETCs issued by US companies has been in effect since 1994. The efficacy of Section 1110 of the US Bankruptcy Code, which US bankruptcy courts have repeatedly upheld, is a critical feature upon which the structure of EETCs is built, typically leading to no downward notching from the preliminary outcome for transactions where the US Bankruptcy Code applies.

For EETCs whose underlying financings are subject to the Cape Town Convention or other non-US insolvency regimes, we consider the extent to which the relevant legal system provides a similar level of certainty around the treatment of EETC obligations as does the US Bankruptcy Code.

The methods of adopting or implementing the selected provisions of the Cape Town Convention in a country's national laws can vary, and there may be few enforcement precedents to demonstrate that compliance with the Cape Town Convention will lead to timely access to collateral, which may result in downward notching for this notching factor even in situations where the Cape Town

Convention has been implemented by the country whose law governs the underlying financing transaction or lease. Precedents of enforcement may inform our expectation of bankruptcy outcomes in a particular jurisdiction, but this experience would not necessarily be relevant to our views of the bankruptcy regime in other jurisdictions.

A country's historical compliance with international treaties and its domestic laws, including in relation to an owner's rights to repossess assets following a lessee default, is a key consideration in our assessment of the legal framework for EETCs in non-US jurisdictions. The institutions and governance strength of the country of domicile of non-US airline EETC issuers, determined during our sovereign rating assessment, including the quality of institutions, rule of law and control of corruption, also informs our assessments.

Downward notching may result, for example, from an EETC being subject to a legal jurisdiction where Cape Town Convention implementing legislation has not been put in place, or where there is a risk that the Convention may not be recognized by the courts or may be applied in a way that reduces EETC creditor protections.

**How we assess it for ETCs**

*Cross-default, cross-collateralization and cross-subordination*

These features do not apply to ETC structures as they are typically structured with a single tranche of notes.

*Liquidity*

We do not notch down for the lack of a liquidity facility or PIK feature as these are not part of an ETC transaction structure. The higher probability of default of an ETC relative to an EETC is captured in the LTV grid.

*Legal framework*

Rolling stock ETCs are subject to Section 1168 of the US Bankruptcy Code, the terms of which emulate Section 1110 applicable to aircraft. Given that US courts have repeatedly upheld the efficacy of Section 1110, we expect they would similarly uphold the efficacy of Section 1168. A change in our view of the efficacy of Section 1168 may result in downward notching for this notching factor.

## Other considerations

Ratings may reflect consideration of additional factors that are not in the scorecard, usually because the factor's credit importance varies widely among the issuers in the sector or because the factor may be important only under certain circumstances or for a subset of issuers. Such factors can include financial controls and the quality of financial reporting; legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends, can also affect ratings.

### Impact of minor or temporary changes in LTV

Minor or temporary changes in our estimated collateral valuations may result in changes to the indicated notching in our LTV grids. For example, if the LTV of a senior instrument with a Baa3 airline changes to 65.1% from 64.9%, the indicated notches of uplift from our LTV grid would decrease to five from six. Where we consider such notching changes to be minor or temporary, they typically do not affect our assigned ratings.

Similarly, as transactions near their final maturity, the outstanding debt amount may, in some cases, reduce more quickly than our estimated valuation of the corresponding collateral does, which may result in a reduction in the LTV and an increase in the number of notches of uplift indicated by our LTV grids. We generally do not reflect anticipated LTV changes as transactions near maturity in our assigned ratings.

### Foreign-exchange risk for foreign-currency-denominated certificates

Aircraft sales are typically transacted in US dollars. Therefore, for non-US dollar-denominated certificates, we consider the effect of changes in exchange rates between the currency in which an EETC transaction is denominated and the US dollar, as well as the use of financial instruments or derivatives to mitigate foreign-exchange risk. We consider the historical and forecast volatility of the relevant currency exchange rate and the impact a sustained depreciation of the US dollar may have on the LTV ratio and excess collateralization.

### Spare parts and aircraft engine EETCs

Spare parts and aircraft engine EETCs are typically issued in single-note, single-mortgage structures, and the airline must fully affirm or disaffirm the underlying note obligations following a bankruptcy, i.e., it may not selectively affirm obligations related to only some spare parts or engines. We consider the likelihood of affirmation to be generally higher for spare parts and engine EETCs than for aircraft EETCs because airlines need to maintain access to an adequate supply of spare parts and spare engines. Losing access to spare parts could severely affect an airline's operations, potentially resulting in suspension or revocation of the airline's operating certificate by aviation authorities. Also, transaction documents for spare parts or engine EETCs typically stipulate a ceiling for the LTV. Semiannual appraisals of the spare parts inventory or the engines are typically required as well. There is potential for significant swings in value between an engine that has recently been serviced and one that is due for maintenance service. The terms of these transactions require a top-up in the form of a paydown of debt, a cash contribution, the addition of additional spare parts or engines or other collateral, or a combination of these to maintain the certificate LTV ratio at or below required maintenance levels.

An airline would have great difficulty maintaining its operations if it lost access to its spare parts inventory. However, under a disaffirmation and Chapter 7 liquidation, there would be greater risk of loss for a spare parts EETC than for an aircraft EETC. There is greater potential for shrinkage or other value impairment on liquidations of spare parts, unless a high proportion of the spare parts includes rotables, i.e., aircraft parts that can be repeatedly repaired, restoring their functionality to like-new condition, for in-demand aircraft models and engines. Avionics, landing gear and certain engine components are examples of rotables.

### Changes in typical deal structures

While EETCs and ETCs have typically been structured in a relatively similar pattern in recent years, transactions could include different structural features or a combination of features that would affect our analysis and ratings. Structures might also change in response to changes in regulations and tax laws. In assessing the impact of different structural features on default probabilities and loss given default, typical considerations would likely include the way structures allocate risk, their impact on the rights and obligations of transaction parties, and our confidence level in how such structures would fare in stress and distress scenarios, which may be informed by legal opinions or other third-party opinions or assurances.

## Using the scorecard to arrive at a scorecard-indicated outcome

### 1. Measurement or estimation of factors in the scorecard

In the "Discussion of the scorecard factors" section, we explain our analytical approach for scoring each scorecard factor, and we describe why they are meaningful as credit indicators. We score at the factor level, as the scorecard factors do not have sub-factors.

The information used in assessing the factors is generally found in or calculated from information in the issuer offering memoranda, third-party aircraft valuations and obligor financial statements, derived from other observations or estimated by Moody's analysts. We may also incorporate non-public information.

Our ratings are forward-looking and reflect our estimates of future collateral values and financial and operating performance. However, historical results and collateral values are helpful in understanding patterns and trends of an EETC or ETC's performance, as well as for peer comparisons.

### 2. Determining the overall scorecard-indicated outcome

For each instrument, the airline or railroad CFR, issuer rating or senior unsecured rating is notched upward based on the instrument's seniority and the scoring from the applicable LTV grid to arrive at an alphanumeric score before the notching factors (the preliminary outcome).

We then consider whether the preliminary outcome should be notched upward or downward for the qualitative collateral attributes or financing terms notching factors. In aggregate, the notching factors can result in a total of up to one upward notch or up to four downward notches from the preliminary outcome to arrive at the scorecard-indicated outcome.

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we assign individual instrument ratings.

## Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, please see *Rating Symbols and Definitions*.[2]

## Limitations

In the preceding sections, we have discussed the factors and many of the other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the overall rating methodology.

### General limitations of the methodology

This methodology document does not include an exhaustive description of all factors that we may consider in assigning ratings to EETCs and ETCs. EETCs and ETCs may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations into ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a financing's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, regulatory and legal actions or force majeure events. In any case, predicting the future is subject to substantial uncertainty.

## Moody's related publications

Credit ratings are primarily determined by sector credit rating methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. An index of sector and cross-sector methodologies can be found here.

For data summarizing the historical robustness and predictive power of credit ratings, please click here.

For further information, please refer to *Rating Symbols and Definitions*, which is available here.

*Moody's Basic Definitions for Credit Statistics (User's Guide)* can be found here.

## Endnotes

1 The Convention on International Interests in Mobile Equipment is an international treaty established in October 2001 in Cape Town, South Africa that had been enacted by 80 countries and the EU by 2022. This convention was created to promote the financing or leasing of movable equipment, specifically, aircraft including helicopters, aircraft engines, railroad rolling stock including locomotives, and space assets, including rockets and satellites. A protocol to the convention on matters specific to aircraft equipment was also created to guide the implementation of the convention by adopting states. The convention and protocol are collectively referred to as the Cape Town Convention.

2 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

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